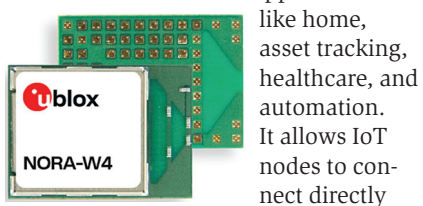


to be beneficial, especially with higher stacks, as it helps mitigate chip die warping associated with thinner die. It offers over 20% enhanced vertical density compared to its other product. Its performance and capacity will enable more flexible resource management and lower data centre TCO. In AI applications, it can boost AI training speed by 34% and increase inference service users by over 11.5 times compared to HBM3 8H.

Samsung Electronics
<https://www.samsung.com>

Compact and versatile wi-fi 6 module with enhanced security

Recently, u-blox has launched the NORA-W4 module, featuring wireless technologies such as Wi-Fi 6, Bluetooth LE 5.3, thread, and Zigbee. Its size (10.4mm x 14.3mm x 1.9 mm) and cost make it suitable for IoT applications



like home, asset tracking, healthcare, and automation. It allows IoT nodes to connect directly to Wi-Fi, eliminating the need for a Bluetooth gateway and reducing system costs. This feature is beneficial for wireless sensors. With Wi-Fi 6 technology optimised for the IoT, the module minimises congestion in settings such as factories and warehouses, enhancing throughput and reducing latency.

u-blox
<https://www.u-blox.com>

Analogue switches to protect 1.8V electronic systems

Nexperia has launched a new line of 4- and 8-channel analogue switches designed for monitoring and protecting 1.8V electronic systems. The NMUX130X series includes multiplexers with AEC-Q100 qualified variants for automotive use and standard versions for applications such as sensor monitoring, diagnostics, enterprise computing, and

household appliances. The switches have control pins that function independently of their power supply voltage



range, eliminating the need for extra components for level-shifting. The fail-safe logic enables control pins to be pre-biased before system power-up and maintain bias post-power-down, eliminating the need for power sequencing with the system MCU. The devices incorporate injection current control circuitry, limiting active channel voltage shifts to under 5mV during overvoltage events on disabled channels, thereby removing the necessity for external diode and resistor networks, saving board space, and reducing costs.

Nexperia
<https://www.nexperia.com>

World's smallest low-power UWB SoC

Shenzhen Giant Microelectronics Company Limited (GiantSemi) has introduced the GT1500 ultra-wideband (UWB) SoC, an industry-leading chip for



ranging, positioning, and wireless connectivity. The SoC stands out for its low power consumption and high performance. It is the world's smallest UWB single-chip SoC, housed in a compact package with a minimal footprint of 9mm². It integrates four receiver channels, RF, analogue, and baseband functions, along with an embedded MCU for control and protocol processing, all in one device. The SoC is suitable for wearable and IoT products due to its small size, which reduces design complexity, component count, and costs.

GiantSemi
<https://www.giantsemi.com>

World's most advanced 5G modem-RF system

Qualcomm Technologies, Inc., has announced the Snapdragon X80 5G modem-RF system, its seventh genera-

tion 5G modem-to-antenna solution. It features a dedicated 5G AI processor and is ready for 5G-advanced architecture, achieving several world-first milestones such as integrated NB-NTN satellite communications support, a 6-antenna architecture for smartphones, 6X carrier aggregation, and AI-based

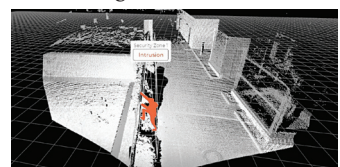


mm wave range extension for fixed wireless access customer premise equipment (CPE). The Snapdragon X80 introduces AI innovations that enhance data speeds, latency, quality of service (QoS), coverage, location accuracy, spectrum efficiency, power efficiency, and multi-antenna management. The 5G modem-RF system empowers OEMs and operators to create next-generation devices supporting 5G Advanced, with unrivalled capabilities and performance.

Qualcomm Technologies
<https://www.qualcomm.com>

Versatile 3D lidar sensor for security purposes

Blickfeld has launched the 'Blickfeld QbProtect,' a 3D lidar sensor for security applications. It has on-device software for monitoring zones, fences, and



facades, processing data on unauthorised entry without extra computers and providing results in common data formats. The sensor ensures object detection under various conditions, with a low false alarm rate and security. It can be installed, integrated into security systems, and configured through a graphical user interface. Through a 3D viewer in the web-based interface, users can define security zones using drag-and-drop and set the object size for triggering alarms. This reduces false alarms by excluding small animals or vegetation from security warnings.

Blickfeld
<https://www.blickfeld.com>